TECH MAVERICKS

GREENROUTE AI

AI-Powered Eco-Friendly Delivery & Route Optimization System

Transportation & Logistics / Sustainability

Clean Routes. Fewer Trips. Lower Emissions.



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Meet Our Team

TECH MAVERICKS

S

Surabhi

Team member

M

Miruthula

Team member

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Samrakshana

Team member

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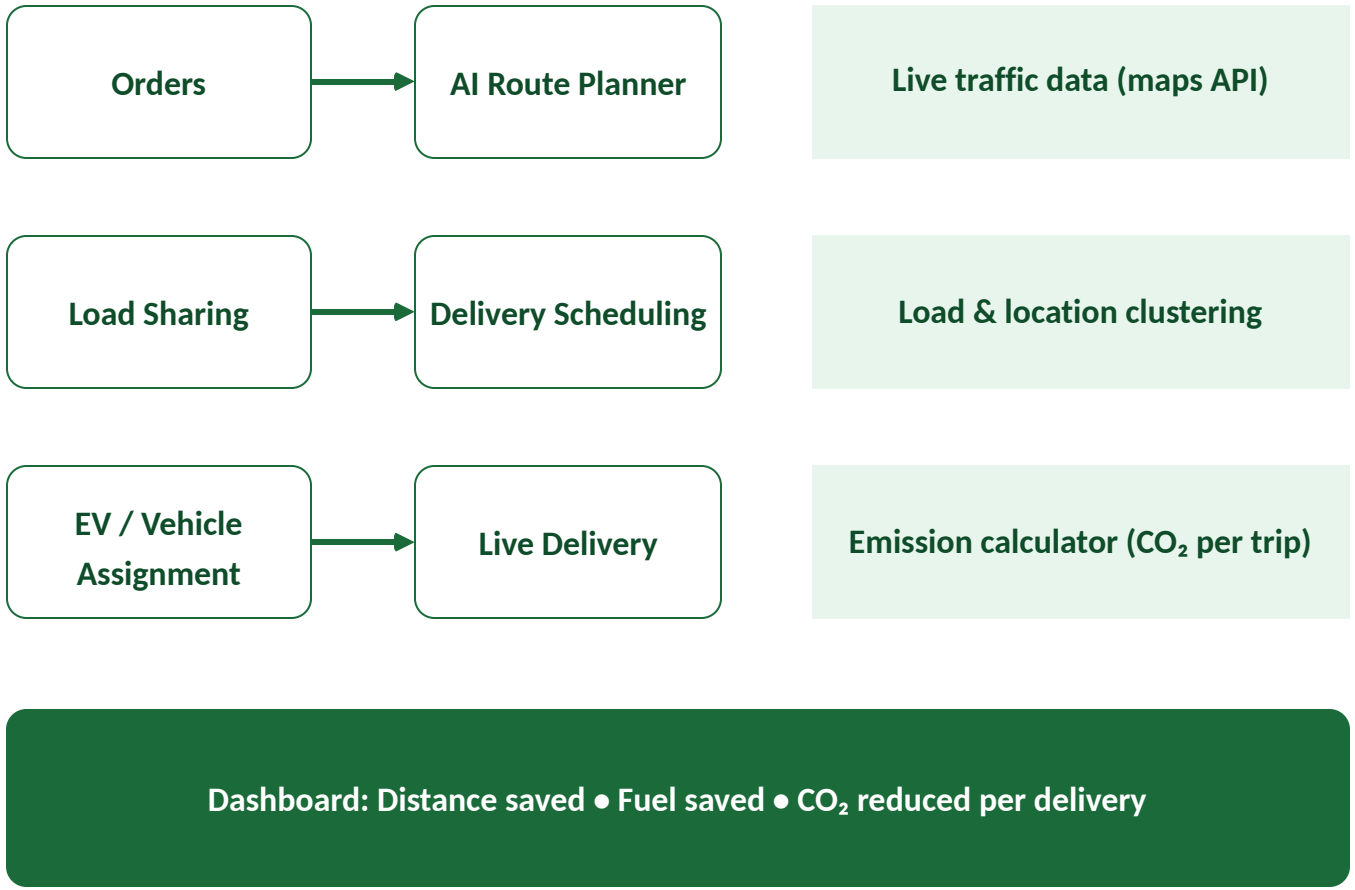
GREENROUTE AI | PROPOSED SOLUTION

Problem

- Delivery vehicles burn more fuel and release harmful gases.
- Traffic jams raise delivery time and fuel use.
- Empty-space trips and unnecessary runs raise cost and pollution.

Solution

- Smart Route Planning: AI picks the shortest practical, low-traffic route.
- Eco-Friendly Vehicles: EVs for local deliveries.
- Load Sharing: combine nearby deliveries in one vehicle.
- Carbon Emission Tracking per delivery.
- Smart Scheduling: group deliveries by location.



Uniqueness: one platform that combines routing, load sharing, EV use and emission tracking.

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GREENROUTE AI | TECHNICAL APPROACH

Data & Hardware

- GPS / vehicle tracking
- Live traffic & map data
- Order location & parcel size
- EV battery / fuel data
- Mobile device for drivers

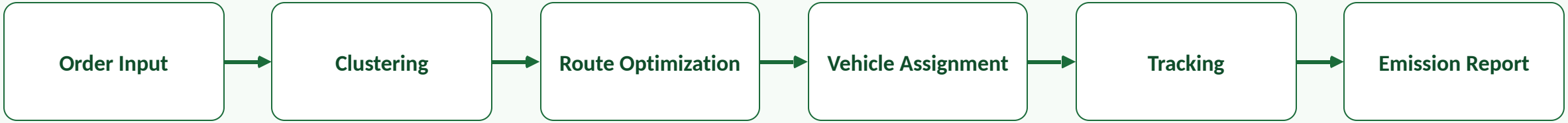
Software

- Python, REST APIs
- Maps / routing API (OSRM or Google)
- Database: PostgreSQL / Firebase
- Web dashboard + driver app

AI / ML Models

- Shortest-route optimization (VRP)
- Delivery clustering by location
- Traffic-time prediction
- Emission estimation per trip

SYSTEM FLOW



Ensures: accurate routing • less fuel use • real-time data • scalable design

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GREENROUTE AI | IMPACT & BENEFITS

Environmental Impact

- Lower fuel use and fewer harmful gases.
- Per-delivery CO_2 tracking builds awareness.
- Promotes electric vehicles for local delivery.

Economic Impact

- Lower transport and fuel cost.
- Fewer empty and unnecessary trips.
- Better use of each vehicle's capacity.

Operational Benefits

- Faster, more on-time deliveries.
- Less time lost in traffic.
- Clear dashboard for planning and reports.

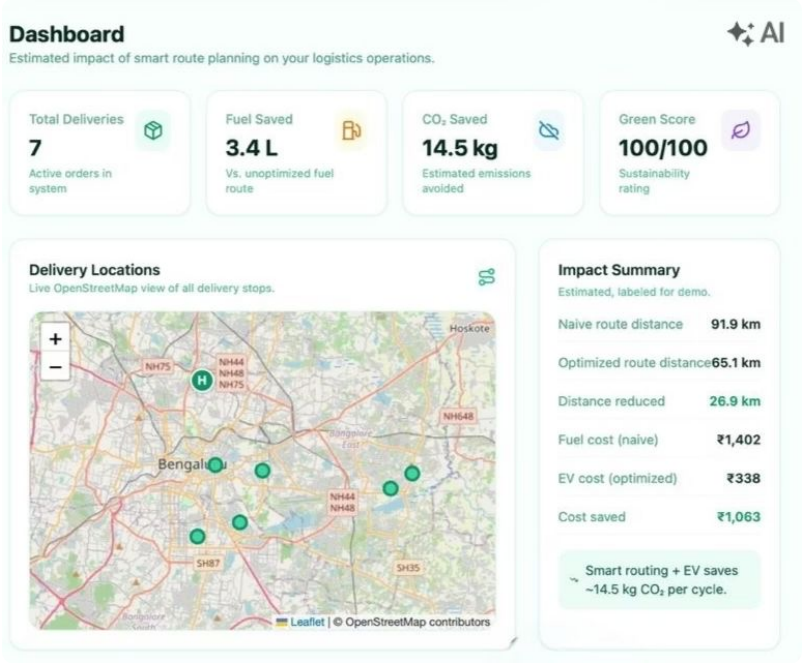
Future Vision & Scalability

- Add AI demand forecasting.
- Integrate with e-commerce and city logistics.
- Expand to EV fleets and shared hubs.

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Our Web Application – GreenRoute AI

Smart Route Planning and Environmental Impact Dashboard



Our GreenRoute AI web application demonstrates smart delivery route planning, distance optimisation, estimated fuel savings, and carbon-emission tracking to support greener logistics.

Web Application Prototype – Tech Mavericks

Dashboard figures are demonstration estimates, not verified real-world results.

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